

CODE-MIXED HATE SPEECH DETECTION IN SINHALA AND TAMIL USING MACHINE LEARNING AND NLP TECHNIQUES

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DECLARATION

We hereby certify that this dissertation represents our own work, which has been performed after registration for the degree of Bachelor of Information and Communication Technology Honours Degree in the Department of Information and Communication Technology at the Faculty of Technology, University of Sri Jayewardenepura, Sri Lanka. To the best of our knowledge and belief, it does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university, and does not contain any material previously published or written by another person except where due reference is made in the text.

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ABSTRACT

CHAPTER 01: INTRODUCTION AND BACKGROUND

1.1 Overview of the Proposed Study

Hate speech on social media has emerged as a significant societal concern in Sri Lanka, a country characterized by linguistic diversity, historical ethnic sensitivities, and rapid digital transformation. With over 12.4 million internet users and social media penetration exceeding 35% of the population in recent years, platforms such as Facebook, Instagram, and X have become central arenas for public discourse. While these platforms enable communication and civic participation, they also facilitate the rapid spread of harmful and divisive content.

Sri Lanka's multilingual environment adds an additional layer of complexity. Users frequently engage in code-mixed communication, blending Sinhala, Tamil, and English within a single sentence or post. This linguistic blending reflects natural conversational practices but presents serious challenges for automated hate speech detection systems. Code-mixed text often includes Romanized transliterations, Unicode script mixing, slang, sarcasm, abbreviations, emojis, and implicit expressions. These characteristics make detection particularly difficult for conventional moderation tools, most of which are designed for monolingual, high-resource languages such as English.

Existing automated systems frequently exhibit high false-negative rates when applied to low-resource and code-mixed languages, allowing harmful content to remain undetected. In a context where ethnic and religious tensions have historically contributed to social instability, ineffective moderation may amplify societal divisions.

This dissertation proposes the development of a robust, context-aware hate speech detection

system specifically designed for Sinhala–Tamil–English code-mixed social media text. The study adopts a structured three-tier experimental framework:

1. Establishing baselines using traditional feature engineering and classical machine learning models.
2. Implementing hybrid models that combine multilingual sentence embeddings with cost-sensitive classifiers.
3. Developing a transformer-based framework (CM-SCALE) incorporating code-mix-aware adaptations and parameter-efficient fine-tuning strategies.

The research integrates existing resources such as the Sinhala Offensive Language Dataset (SOLD) and public Dravidian code-mixed corpora, while constructing a unified and stratified dataset of approximately 47,000 annotated samples. Rigorous preprocessing is applied, including Romanized normalization, script handling, and emoji/hashtag preservation. Performance is evaluated using macro-F1, precision, recall, and accuracy, alongside detailed error analysis.

By addressing linguistic, cultural, and computational challenges unique to Sri Lanka’s digital ecosystem, this study aims to advance multilingual NLP research and contribute to safer online spaces in low-resource multilingual societies.

1.2 Research Problem

Despite the growing prevalence of hate speech in Sri Lanka’s multilingual social media landscape, existing detection mechanisms remain insufficient for Sinhala–Tamil–English code-mixed content. The core research problem lies in the inadequacy of current models to effectively process and classify linguistically complex, low-resource, and mixed-script inputs.

Several interconnected challenges contribute to this issue. First, there is a scarcity of large-scale, high-quality annotated datasets specifically tailored for Sinhala–Tamil–English code-mixed hate speech. Second, linguistic complexities such as sarcasm, implicit targeting, phonetic transliterations, emojis, and informal slang introduce ambiguity that traditional models struggle to interpret. Third, monolingual models trained on high-resource languages exhibit poor generalization when applied to low-resource multilingual environments, leading to elevated false-negative rates.

Furthermore, most existing research addresses Sinhala and Tamil independently or focuses solely on Tamil-English code-mixing, leaving a critical gap in unified modeling approaches capable of handling Indo-Aryan (Sinhala), Dravidian (Tamil), and English simultaneously within real-world Sri Lankan discourse.

The absence of comprehensive, code-mix-aware detection frameworks allows harmful content targeting ethnic, religious, or community groups to spread rapidly online, potentially intensifying offline tensions. Addressing this gap requires not only improved modeling techniques but also structured experimentation and real-world validation.

1.3 Research Objectives

1.3.1 Main Objective

To develop a robust and efficient hate speech detection system capable of accurately classifying Sinhala–Tamil–English code-mixed social media text using progressive machine learning and transformer-based natural language processing techniques.

1.3.2 Specific Objectives

This study seeks to achieve the following specific objectives:

- To construct, curate, and preprocess a unified dataset of approximately 47,000 code-mixed samples sourced from publicly available social media content and benchmark datasets, ensuring ethical compliance and annotation reliability.
- To implement and evaluate baseline models using traditional feature engineering techniques such as TF-IDF and classical classifiers including Support Vector Machines and Logistic Regression.
- To develop hybrid models by integrating multilingual sentence embeddings (e.g., LaBSE) with cost-sensitive classification strategies to enhance contextual understanding.
- To design and evaluate a transformer-based framework incorporating parameter-efficient fine-tuning mechanisms and class imbalance mitigation strategies for improved multilingual representation.
- To systematically compare model performance across tiers using macro-F1, precision, recall, accuracy, and confusion matrix analysis, including focused evaluation on high code-mix and emoji-rich samples.
- To conduct detailed error analysis and interpretability assessment in order to identify limitations in sarcasm handling, slang recognition, and implicit hate detection.

1.4 Significance of the Study

This research carries both theoretical and practical significance. From a theoretical perspective, the study advances NLP research for low-resource and code-mixed languages by introducing a structured three-tier evaluation framework tailored to Sinhala–Tamil–English discourse. It

contributes empirical insights into multilingual representation learning, class imbalance mitigation, and the treatment of Romanized transliterations and emoji signals in hate speech detection.

From a practical standpoint, the research delivers a validated hate speech detection model and a curated large-scale dataset that can support moderation efforts in Sri Lanka. The system's deployment within a web-based interface demonstrates real-world feasibility, bridging the gap between academic experimentation and operational implementation.

From a societal perspective, the project contributes to fostering safer digital communication spaces. By improving detection of content targeting ethnic and religious groups, the study supports social cohesion, responsible freedom of expression, and ethical AI development in multilingual societies.

1.5 Scope and Limitations

Scope

This study focuses on binary classification of text-based social media content (HATE vs. NON-HATE) in Sinhala–Tamil–English code-mixed form. Data sources are restricted to publicly available posts and comments from platforms such as Facebook and Instagram, supplemented by established datasets including SOLD and Dravidian corpora.

Experiments are conducted using open-source libraries such as scikit-learn and transformer-based frameworks within accessible computational environments. Evaluation emphasizes macro-F1 and balanced performance across classes, particularly in code-mixed subsets.

Limitations

The study is limited to textual modality and does not incorporate multimodal analysis (images, videos, or audio) due to time and resource constraints. Although the dataset size of approximately 47,000 samples is substantial for a low-resource context, it remains moderate relative to large-scale multilingual corpora, which may influence deep model generalization.

The models may underperform on rare dialectal expressions, highly obfuscated slang, or newly emerging evasion techniques absent from training data. Additionally, while a web-based interface is implemented for demonstration, full-scale API integration with commercial social media platforms is beyond the scope of this dissertation.

Ethical constraints restrict data collection to publicly accessible sources; private, deleted, or restricted content is excluded.

1.6 Expected Outcomes

The anticipated outcomes of this research include:

- A curated and stratified dataset of approximately 47,000 annotated Sinhala–Tamil–English code-mixed samples.
- A structured three-tier comparative modeling analysis.
- A transformer-based model achieving the highest performance, with observed accuracy of approximately 0.84 and strong macro-F1 balance.
- A functional web-based hate speech detection interface demonstrating real-time usability.
- A reproducible framework adaptable to other low-resource multilingual environments.

CHAPTER 02: LITERATURE REVIEW

2.1 Introduction

Hate speech detection has emerged as a critical research domain within Natural Language Processing (NLP), primarily due to the exponential growth of user-generated content on social media platforms. The rapid expansion of digital communication has intensified the need for automated moderation systems capable of identifying harmful, abusive, or discriminatory content at scale. While substantial advancements have been achieved in high-resource languages such as English, the research landscape for low-resource and multilingual environments—particularly Sinhala and Tamil—remains comparatively underdeveloped.

The complexity increases further in multilingual societies like Sri Lanka, where code-mixing between Sinhala, Tamil, and English is common in online discourse. Users frequently alternate between scripts, employ Romanized spellings, insert emojis, and use informal abbreviations, thereby creating highly unstructured textual data. These characteristics introduce significant computational and linguistic challenges.

This chapter critically examines prior research relevant to Sinhala, Tamil, and multilingual code-mixed hate speech detection. It traces the methodological evolution from traditional machine learning to transformer-based architectures, evaluates strengths and weaknesses of each approach, and identifies research gaps that motivate the proposed CM-SCALE framework.

2.2 Evolution of Hate Speech Detection Approaches

The methodological progression of hate speech detection research can broadly be categorized into three stages:

1. Traditional Machine Learning Approaches
2. Deep Learning Models (Non-Transformer Architectures)
3. Transformer-Based and Hybrid Systems

This evolution reflects increasing model sophistication and improvements in contextual understanding. Early models relied heavily on handcrafted features and statistical learning techniques, whereas modern systems utilize large-scale pre-trained language representations capable of capturing deep semantic relationships. Each stage contributed valuable insights, yet each also revealed limitations—particularly when applied to low-resource, multilingual, and code-mixed environments.

2.3 Traditional Machine Learning Approaches

Initial studies in hate speech detection employed classical machine learning algorithms such as Support Vector Machines (SVM), Logistic Regression, Naïve Bayes, and Random Forests. These models were typically combined with feature extraction techniques including Term Frequency–Inverse Document Frequency (TF-IDF), bag-of-words representations, and character n-grams.

Research focusing on Sinhala and Tamil languages, including works by Fernando and Deng (2023) and Hettiarachchi et al. (2020), demonstrated that traditional classifiers could achieve moderate accuracy levels, generally ranging between 66% and 71%, particularly when trained on

relatively small datasets. These approaches were computationally efficient, required limited hardware resources, and offered a degree of interpretability that facilitated error analysis.

However, several limitations were consistently observed. Classical models struggle to capture contextual semantics beyond surface-level word frequency. They are highly sensitive to spelling variations, which are common in Romanized Sinhala and Tamil. Additionally, these models exhibit weak performance when confronted with sarcasm, implicit hate, or nuanced targeting language. Their inability to generalize effectively to heavily code-mixed inputs further restricts their applicability in real-world Sri Lankan social media contexts.

Despite these shortcomings, traditional machine learning methods remain valuable as strong baseline systems. They provide essential comparative benchmarks and perform competitively when handling high-dimensional sparse feature representations.

Critical Insight:

While traditional models offer efficiency and interpretability, their shallow representation mechanisms limit their effectiveness in multilingual and context-rich hate speech detection scenarios.

2.4 Deep Learning Approaches (Non-Transformer Models)

The limitations of classical approaches motivated the adoption of neural network-based models capable of capturing sequential dependencies. Architectures such as Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), Bidirectional LSTM (BiLSTM), Gated Recurrent Units (GRU), and Hierarchical Attention Networks (HAN) were introduced into hate speech detection research.

Studies such as Shibly et al. (2021) and Samarasinghe et al. (2020) applied LSTM-based architectures to Sinhala datasets and reported notable improvements in contextual modeling compared to traditional machine learning methods. Similarly, hierarchical attention mechanisms were explored for Tamil-English code-mixed data to better capture phrase-level and sentence-level importance patterns.

These models demonstrated improved F1-scores—reaching as high as 0.80 to 0.93 in certain tasks—indicating enhanced semantic modeling and contextual awareness. Neural architectures were more capable of identifying patterns in sequences and modeling dependencies across words.

Nevertheless, deep learning models introduced new challenges. They often relied on static or task-specific embeddings that lacked strong cross-lingual transfer capability. Their performance deteriorated when confronted with extreme code-mixing or low-resource training data. Additionally, neural networks demanded significantly higher computational resources relative to dataset size, making experimentation more expensive and time-consuming.

Critical Insight:

Although deep learning approaches enhanced contextual sensitivity, their dependence on limited embedding representations and reduced multilingual transfer capability constrained their scalability in low-resource code-mixed environments.

2.5 Transformer-Based Models

The introduction of transformer architectures marked a transformative shift in NLP research. Pre-trained multilingual models such as XLM-RoBERTa (XLM-R), mBERT, and MuRIL

leverage large-scale multilingual corpora to generate contextual embeddings capable of modeling semantic relationships across languages.

Recent studies in Indic and Dravidian language contexts have demonstrated significant performance improvements using transformer fine-tuning strategies. In certain Tamil hate speech detection tasks, macro-F1 scores exceeding 0.85 have been reported, underscoring the superiority of contextualized embeddings over static representations.

Transformer-based models offer several key advantages:

- Deep contextual word representations
- Cross-lingual transfer learning
- Robust handling of semantic nuance
- Improved performance in low-resource languages

However, research gaps persist. Many studies focus exclusively on either Sinhala-English or Tamil-English code-mixing, without attempting a unified modeling strategy across all three languages. Furthermore, limited attention has been given to preserving emojis and hashtags, which often carry implicit aggression signals. Structured comparative evaluations across modeling paradigms are also scarce, and deployment-focused validation is rarely addressed.

Transformer models are computationally intensive and sensitive to hyperparameter configurations. Without systematic experimentation and validation strategies, reported improvements may lack reproducibility.

Critical Insight:

Transformers represent the current state-of-the-art for multilingual hate speech detection, but require careful adaptation, structured comparison, and deployment-oriented validation in code-mixed Sri Lankan contexts.

2.6 Code-Mixed Language Challenges

Code-mixing involves the blending of multiple languages within a single sentence or discourse segment. In Sri Lankan digital communication, Sinhala-English and Tamil-English mixtures are highly prevalent. Users frequently alternate scripts, employ phonetic transliterations, and integrate emojis to convey sentiment.

This creates multiple computational challenges:

- Mixed scripts (Unicode and Romanized forms)
- Spelling inconsistencies
- Informal abbreviations
- Implicit hate disguised as humor or sarcasm
- Emoji-based sentiment reinforcement

Most existing research addresses Tamil-English or Sinhala-English independently, but very few studies attempt a unified Sinhala–Tamil–English detection framework. Moreover, many available datasets are relatively small, platform-specific, imbalanced, and limited in annotation transparency. This fragmentation restricts generalizability and reduces cross-lingual robustness.

2.7 Class Imbalance and Evaluation Issues

Hate speech detection datasets are typically imbalanced, with non-hate content significantly outnumbering hate instances. Several studies report high overall accuracy while neglecting balanced evaluation metrics. This can result in misleading performance interpretations.

Common shortcomings include over-reliance on accuracy, insufficient macro-F1 reporting, limited statistical comparison between models, and minimal error analysis discussion. Although some recent works introduce cost-sensitive learning and focal loss techniques, systematic imbalance mitigation strategies are not consistently applied in Sinhala–Tamil research contexts.

2.8 Deployment and Real-World Applicability

A major limitation in existing literature is the lack of deployment validation. While experimental results are often promising, few studies demonstrate end-to-end implementation or real-time system integration. The absence of user interface validation, latency testing, and usability assessment creates a disconnect between academic experimentation and practical applicability.

This gap highlights the importance of evaluating not only model performance metrics but also operational feasibility in real-world environments.

2.9 Identified Research Gaps

Based on the reviewed literature, the following research gaps are identified:

- Absence of a unified large-scale Sinhala–Tamil–English dataset
- Limited structured comparison across classical, hybrid, and transformer-based models
- Inadequate preservation and analysis of emojis and Romanized normalization

- Insufficient systematic class imbalance mitigation
- Lack of progressive tier-based evaluation strategies
- Minimal deployment validation and usability testing

These gaps justify the need for a structured and comprehensive framework.

2.10 Positioning of the Proposed Study

The proposed CM-SCALE framework directly addresses these limitations by constructing a stratified dataset of approximately 47,000 samples integrating Sinhala, Tamil, and English content. Unlike many prior studies, emojis and hashtags are preserved during preprocessing, and transliteration normalization is applied to Romanized segments to reduce spelling variability.

The framework introduces a structured three-tier modeling strategy, systematically progressing from classical machine learning to hybrid embedding-based models and ultimately to transformer fine-tuning. Models are evaluated using accuracy, precision, recall, F1-score, and macro-F1 to ensure balanced assessment.

Finally, the best-performing model is deployed within a functional web interface, bridging the gap between theoretical experimentation and practical application. This dual emphasis on methodological rigor and real-world validation ensures both academic contribution and societal relevance.

CHAPTER 03: METHODOLOGY

3.1 Research Design

This study adopts an applied, experimental, and comparative research design with the primary objective of developing and rigorously evaluating a practical hate speech detection system for Sinhala–Tamil–English code-mixed social media text. Unlike purely theoretical research, this study is solution-oriented and focuses on designing, implementing, and validating a deployable system within a real-world multilingual context.

The research follows a structured experimental pipeline beginning with dataset construction and preprocessing, followed by baseline model implementation, progressive enhancement across three modeling tiers, systematic performance comparison, and final deployment validation. The design intentionally incorporates a comparative modeling strategy to evaluate how performance evolves from traditional sparse feature-based classifiers to hybrid embedding-based approaches and ultimately to contextual transformer architectures.

This progressive evaluation strategy forms the conceptual foundation of the proposed CM-SCALE framework. Rather than assuming transformer superiority, the study empirically validates each modeling stage under identical dataset conditions. This ensures that observed improvements are attributable to model capability rather than dataset variation, thereby strengthening experimental integrity and reproducibility.

3.2 Study Area and Data Sources

The study focuses on Sri Lankan digital discourse, particularly user-generated content published on social media platforms where multilingual communication is common. Sri Lanka presents a unique linguistic environment where Sinhala, Tamil, and English coexist in both formal and informal communication. On social media platforms, users frequently engage in code-mixing by blending these languages within single sentences, often switching between Unicode scripts and Romanized transliterations.

Data were collected from publicly accessible social media content, including Instagram comments and Facebook public posts and comment threads. Additionally, the study integrates established benchmark datasets such as the SOLD (Sinhala Offensive Language Dataset) and publicly available Dravidian code-mixed datasets for Tamil-English content.

The collected data reflect real-world linguistic characteristics, including:

- Mixed script usage (Sinhala/Tamil Unicode combined with Romanized English)
- Phonetic spellings and transliterations
- Informal slang and abbreviations
- Emoji-based sentiment indicators
- Hashtags that amplify contextual meaning

Only publicly available content was used in order to maintain ethical compliance and avoid privacy violations. No private accounts, direct messages, or restricted content were accessed during the data collection process.

3.3 Dataset Construction

3.3.1 Dataset Composition

The final unified dataset consists of approximately 47,000 manually labeled text samples. This represents one of the larger consolidated datasets for Sinhala–Tamil–English hate speech detection in a code-mixed environment.

The dataset integrates multiple sources to ensure diversity and generalizability:

- Sinhala offensive and hate speech samples from the SOLD dataset
- Tamil-English code-mixed hate speech datasets
- Custom-collected Instagram public comments
- Public Facebook comment samples

By combining multiple sources, the dataset captures varied linguistic patterns, writing styles, and platform-specific discourse behaviors. This integration reduces platform bias and improves model robustness.

3.3.2 Data Splitting Strategy

To ensure unbiased evaluation, the dataset was divided using a stratified sampling strategy into three subsets:

- 70% Training set
- 15% Validation set
- 15% Test set

Stratification ensures proportional representation of both HATE and NON_HATE classes across all splits. This is particularly important in hate speech detection tasks where class imbalance can distort model learning and evaluation results.

The validation set was used exclusively for hyperparameter tuning and early stopping decisions, while the test set was held out and used only for final performance evaluation to prevent data leakage.

3.3.3 Label Schema

The study adopts a binary classification schema consisting of two labels:

- HATE
- NON_HATE

Although hate speech can be categorized into multiple subtypes (e.g., abusive, offensive, inciting violence), a binary classification framework was selected to enhance annotation reliability and reduce ambiguity. Multi-class annotation in low-resource, code-mixed contexts often results in inconsistent labeling due to subjective interpretation. Therefore, binary categorization improves consistency and strengthens inter-annotator agreement.

3.4 Annotation Process

Custom-collected data underwent manual annotation following a structured protocol. Two independent annotators were assigned to label the dataset. To ensure annotation quality and reliability, approximately 10–15% of the samples were overlapped between annotators. Disagreements were resolved through discussion and adjudication by a third reviewer.

The annotation guidelines defined hate speech as content that:

- Explicitly targets ethnic, religious, or social groups
- Contains severe abusive or degrading language
- Promotes discrimination or hostility
- Encourages violence or exclusion

Implicit hate speech was also considered when contextual indicators suggested harmful intent. However, ambiguous sarcasm or mild profanity without clear group targeting was labeled as NON_HATE to maintain consistency.

Inter-annotator reliability was measured using Cohen's Kappa, with a target agreement level greater than 0.70, indicating substantial agreement.

3.5 Ethical Considerations

The study adhered to ethical data usage principles. Only publicly available content was collected, and no direct interaction with users occurred. Personally identifiable information (PII), usernames, URLs, and other identifying markers were removed during preprocessing.

The dataset was used exclusively for academic research purposes. Since no human subjects were directly involved and no private data were accessed, formal Institutional Review Board (IRB) approval was not required under institutional guidelines.

3.6 Data Preprocessing

Preprocessing plays a critical role in code-mixed multilingual environments due to linguistic variability and informal writing patterns.

3.6.1 Text Cleaning

The preprocessing pipeline included removal of URLs, excessive punctuation, and irrelevant special characters. Romanized text segments were converted to lowercase to ensure consistency in vectorization. Whitespace normalization was applied to eliminate redundant spacing.

These steps reduce noise while preserving meaningful linguistic content.

3.6.2 Emoji and Hashtag Retention

Unlike many prior studies that remove emojis and hashtags, this study preserves them intentionally. Emojis frequently convey sentiment intensity, including anger, disgust, or aggression. Similarly, hashtags often reinforce contextual hostility or ideological alignment.

For example, angry face emojis or inflammatory hashtags can function as strong predictive features in hate speech classification. Retaining them allows models to capture additional sentiment cues beyond textual content.

3.6.3 Romanized Text Normalization

A significant challenge in Sri Lankan social media text is the widespread use of Romanized Sinhala and Tamil. To address this, transliteration tools were applied to normalize Romanized variants into standardized script representations where possible.

For instance, spelling variations such as “pissu”, “pisu”, and “pissuuwak” were mapped to consistent representations. This reduces sparsity in feature space and improves embedding consistency.

3.6.4 Handling Language Mixing

Although token-level language identification was not explicitly enforced during model training, multilingual transformer architectures inherently capture cross-lingual context through shared subword representations. Therefore, the study relies on model-level multilingual capability rather than explicit segmentation.

3.7 Feature Engineering for Tier 1

For classical machine learning baselines, feature engineering was conducted using both word-level and character-level representations. Word-level TF-IDF vectors capture term importance across documents, while character n-grams (ranging from 3 to 5 characters) help model spelling variations and informal transliterations common in code-mixed text.

Additional features such as emoji frequency counts and hashtag indicators were incorporated to enrich the feature space. These high-dimensional sparse representations are well-suited for linear classifiers such as Logistic Regression and Linear SVM.

3.8 Three-Tier Modeling Framework (CM-SCALE)

The CM-SCALE framework consists of three progressively complex modeling tiers designed to empirically validate performance improvement across methodological advancements.

3.8.1 Tier 1 – Classical Machine Learning Baselines

Tier 1 includes Logistic Regression and Linear Support Vector Machine classifiers with balanced class weights. These models serve as strong baseline systems due to their efficiency, interpretability, and proven effectiveness on TF-IDF features.

Logistic Regression achieved an accuracy of approximately 0.81, while Linear SVM achieved approximately 0.80 accuracy. Although these models demonstrated competitive performance, error analysis revealed limitations in capturing semantic nuance and borderline hate expressions.

3.8.2 Tier 2 – Hybrid Model

Tier 2 integrates pretrained LaBSE sentence embeddings with a tuned Linear SVM classifier. LaBSE, trained on over 100 languages, provides deep multilingual semantic representations capable of handling cross-lingual similarity.

Hyperparameters including kernel selection and regularization strength were optimized using validation data. The hybrid model achieved approximately 0.70 accuracy, with improved recall for hate detection but reduced precision due to sensitivity to borderline semantic expressions.

3.8.3 Tier 3 – Transformer-Based Model

Tier 3 utilizes XLM-RoBERTa (XLM-R) base model fine-tuned for sequence classification. Fine-tuning included addition of a classification head, cross-entropy loss optimization, balanced mini-batches, AdamW optimization, and learning rate scheduling.

Hyperparameters such as maximum sequence length, batch size, and epoch count were experimentally optimized based on validation loss trends. Early stopping was applied to prevent overfitting.

This tier achieved the highest performance with approximately 0.84 accuracy and superior macro-F1, demonstrating strong contextual understanding and robustness to code-mixed input.

3.9 Evaluation Metrics

Performance evaluation was conducted using Accuracy, Precision, Recall, F1-score, and Macro-F1. Given class imbalance concerns, Macro-F1 was prioritized to ensure balanced evaluation across classes. Confusion matrices were generated for each modeling tier to visualize classification distribution and error patterns.

3.10 Class Imbalance Handling

Moderate class imbalance was addressed through stratified sampling, class-weight balancing, balanced mini-batch construction, and threshold tuning. Future improvements may incorporate focal loss to further penalize hard-to-classify samples.

3.11 System Deployment

The best-performing transformer model was integrated into a web-based application enabling real-time prediction. The system workflow consists of user input submission, preprocessing execution, model inference, and prediction output display as HATE or NON_HATE.

System latency and response time were evaluated to ensure practical usability.

3.12 Validation Strategy

To maintain experimental rigor, hyperparameter tuning was performed exclusively on the validation set, and final evaluation was conducted strictly on the held-out test set. No data leakage occurred between dataset partitions, ensuring reliability and reproducibility of results.

CHAPTER 04: RESULT AND DISCUSSION

4.1 Overview of Experimental Evaluation

This chapter presents the preliminary experimental results obtained from the structured three-tier modeling framework described in Chapter 03. Since the research is currently in progress, the results presented here reflect intermediate findings used for the second evaluation milestone.

The evaluation was conducted across three modeling tiers:

- **Tier 1** – Classical Machine Learning Baselines (TF-IDF + Logistic Regression / Linear SVM)
- **Tier 2** – Hybrid Model (LaBSE Embeddings + Linear SVM)

- **Tier 3** – Transformer-Based Fine-Tuned Model (XLM-R based framework)

Performance was measured using Accuracy, Precision, Recall, and Macro-F1 Score. Given the presence of class imbalance and the societal sensitivity of hate speech detection, Macro-F1 was considered the primary evaluation metric.

[Table 4.1: Overall Performance Comparison Across Model Tiers]

4.2 Tier 1 – Classical Baseline Model Results

The baseline models utilized TF-IDF feature extraction combined with Logistic Regression and Linear Support Vector Machines (SVM). These models establish a foundational performance benchmark for Sinhala–Tamil–English code-mixed hate speech detection.

The Logistic Regression model achieved an accuracy of approximately **0.81**, while the Linear SVM demonstrated comparable performance with an accuracy close to **0.80**. Macro-F1 scores indicated relatively balanced classification capability, though slight degradation was observed in the minority hate class.

[Table 4.2: Tier 1 Detailed Metrics – Precision, Recall, Macro-F1]

The confusion matrix analysis reveals that most misclassifications occur when hate speech is expressed implicitly or through sarcasm. Classical models rely heavily on lexical frequency patterns and therefore struggle with contextual nuance, emoji-based signaling, and transliterated variants.

[Figure 4.1: Confusion Matrix – Linear SVM Baseline]

Although Tier 1 models performed reasonably well, their dependence on surface-level textual features limits their ability to capture deeper semantic relationships in highly code-mixed input.

4.3 Tier 2 – Hybrid Model (LaBSE + Linear SVM)

To enhance contextual representation, Tier 2 integrates multilingual sentence embeddings (LaBSE) with a cost-sensitive Linear SVM classifier. This approach aims to capture semantic similarity across Sinhala, Tamil, and English while mitigating class imbalance.

The hybrid model achieved an accuracy of approximately **0.70**. While embedding-based representation improved contextual understanding in some samples, overall performance was slightly lower than classical baselines during this intermediate evaluation stage.

[Table 4.3: Tier 2 Performance Metrics]

The performance variation may be attributed to:

- Incomplete hyperparameter tuning at this stage
- Dataset noise in Romanized transliterations
- Limited domain adaptation of pre-trained multilingual embeddings

However, qualitative analysis suggests improved handling of cross-lingual semantics compared to purely lexical models.

[Figure 4.2: Performance Comparison – Tier 1 vs Tier 2]

4.4 Tier 3 – Transformer-Based Model Results

The transformer-based model built upon XLM-R architecture demonstrated the highest performance among the three tiers during the second evaluation.

The model achieved an accuracy of approximately **0.84**, with improved macro-F1 balance across both hate and non-hate classes. This improvement highlights the effectiveness of contextualized multilingual representations in modeling complex code-mixed discourse.

[Table 4.4: Tier 3 Detailed Evaluation Metrics]

Transformer models leverage deep self-attention mechanisms, enabling better interpretation of:

- Implicit targeting
- Mixed-script expressions
- Emoji-assisted hate signals
- Context-dependent sarcasm

[Figure 4.3: Confusion Matrix – Transformer Model]

The improvement over classical and hybrid approaches indicates that contextual embeddings significantly enhance detection performance in multilingual low-resource environments.

4.5 Comparative Analysis Across Tiers

A comparative analysis of the three tiers demonstrates a progressive improvement from surface-level lexical modeling to deep contextual representation.

[Figure 4.4: Tier-wise Accuracy and Macro-F1 Comparison Bar Chart]

Key observations include:

1. Classical models provide strong baselines but struggle with contextual subtleties.
2. The hybrid approach introduces semantic awareness but requires further tuning.
3. The transformer-based model currently achieves the best overall performance.

Notably, macro-F1 improvements in Tier 3 suggest better balance in detecting minority hate instances, which is critical for ethical AI deployment.

4.6 Error Analysis (Preliminary Observations)

Preliminary qualitative error analysis identified several recurring misclassification categories:

- Sarcastic hate masked as humor
- Indirect ethnic targeting
- Creative spelling variations
- Newly emerging slang terms
- Highly code-mixed short comments

[Table 4.5: Sample Misclassified Instances and Observed Patterns]

These findings indicate the need for:

- Further data augmentation
- Slang normalization strategies
- Enhanced fine-tuning with code-mix-aware preprocessing

4.7 Discussion

The draft experimental results demonstrate that multilingual transformer architectures outperform traditional and hybrid approaches in Sinhala–Tamil–English code-mixed hate speech detection. The progression from TF-IDF-based models to contextualized transformers reveals the importance of semantic representation in low-resource multilingual NLP tasks.

However, the research is ongoing. Hyperparameter optimization, expanded dataset validation, and further ablation studies are required before drawing definitive conclusions. The current findings primarily validate the feasibility of the proposed three-tier framework and justify continued development of the transformer-based system.

Importantly, while numerical improvements are observed, real-world deployment requires careful consideration of ethical implications, bias mitigation, and false-positive control to prevent unjust moderation.